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Evaluation of Few-Shot Learning with Vision Language Models for Needle Bearing Defect Detection

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Overview

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Industrial Artificial Intelligence (IAI)

- Closed loop manufacturing
- Quality assurance, Preventive maintenance, Operational excellence

Needle Bearings

- Great load bearing capacity
- Smaller form factors

Challenges:

- High background noise
- Asymmetric loading
- Limited literature
- CWRU ball bearing dataset

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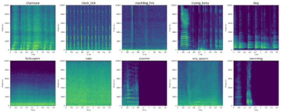
Motivation

Study Setup

- Vision Language Models (VLM) have emerged as a powerful paradigm for multimodal AI
- VLM hase shown remarkable capabilities in wide range of tasks- image captioning, scene description, Visual QandA etc
- Recent VLM models GPT-4o, Claude-Sonnet, Gemini - 1.5Pro demonstrated impressive Zero shot learning capabilties
- How can VLMs infer and analyze spectrograms is an interesting and evolving research topic
- Recent study by Satvik et.al (Nov, 2024) shown VLMs are Few-Shot audio spectrogram classifiers in public

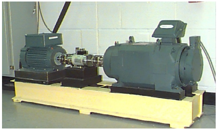


# per class	Selection method	Feature	Accuracy (%)
1	Random	Mel	70.00
1	Hand-picked	Mel	75.00
1	K-means (k=3)	Mel	73.75
1	K-means (k=3)	Amp	60.00
2	K-means (k=3)	Mel	76.25
3	K-means (k=3)	Mel	70.00

Experiment Setup

Study Setup

- Data Collection Setup closely resembles the standard CWRU Bearing Data Center study
- Vibration data is collected using an ICP accelerometer (M353B15)
- Mechanical Load of 50N rotating at constant speed of 900 RPM
- Data Sampled at 24 KHz and collected for 0.6 seconds
- Real time vibration data of needle bearings collected at Schaeffler labs
- Study Span 2 months
- Human Experts generated Ground Truth after manual inspections



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Top 3 Defects of Needle Bearings

Table: Dataset description for the study

Number	Bearing Status	Num Samples
1	No defects found	2800
2	Dents	240
3	Needle Rejects	237
4	Racewy Dents	252

- Top 3 Defects: Dents, Needle Rejects and Raceway Dents
- Downsampled to achieve equal representation of all classes
- For this study 100 samples per class were randomly chosen for testing

Study Methodology

- Ablation study of 2 different spectrogram computations varying parmeters like amplitude scale, frequency axis etc
- Ablation study result better performing spectrogram method is chosen for further analysis
- SOTA model ViT with classifier head is fine tuned and baseline performance captured.
- Human Experts performance of the chosen spectrogram method is taken as baseline scores
- Few shot prompt performance is tested for Top-3 leaderboard VLM namely GPT-4o, Claude-3.5-Sonnet and Gemini-1.5-Pro

STFT Spectrogram

- Short Time Fourier Transform (STFT) - Apply FFT to overlapping signal segments
- Hamming Window used as a Windowing function

$$H(n) = 0.54 + 0.46 \cdot \cos\left(\frac{2\pi}{N} \cdot n\right) \tag{1}$$

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i t \xi} \, dt \tag{2}$$

Parameter	**Parameter values**
Sampling frequency	24 KHz
Spectrogram bandwidth	0-12 KHz
Time samples per window	0.6 seconds
Spectrogram window length	256
Spectrogram hop length	128

Multitaper Spectrogram

- Multi Taper Spectrogram Estimation -
- Orthogonal Slepian Tapers used as Windowing function

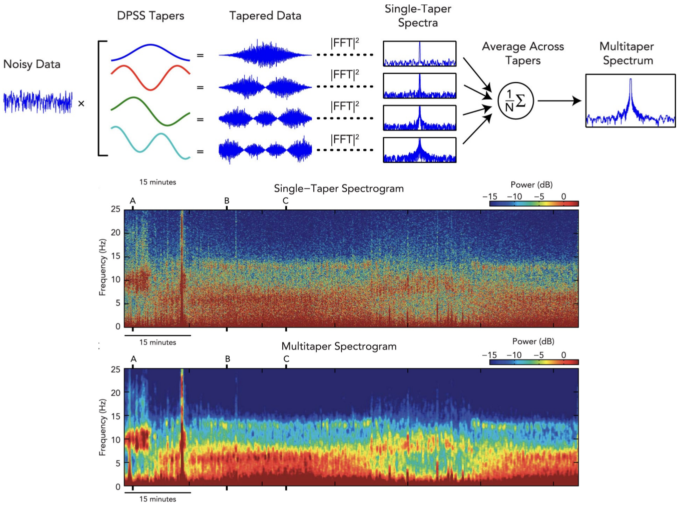
$$TW = \frac{N \cdot \Delta f}{2} \tag{3}$$

$$L \ll \lfloor 2 \cdot TW - 1 \rfloor \tag{4}$$

Table: Multitaper Spectrogram Parameters

Parameter	Parameter values
Sampling frequency	24000 KHz
Time BW Pdt	2.5
Num of Slepian Tapers	4
Spectrogram window length	256
Spectrogram hop length	128

Study Design: Ablation Study MT Estimation Example



Algorithm Run Setup: Fine Tuning ViT Model

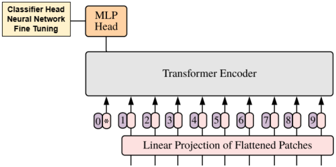


Figure: Fine Tuning ViT model

Table: ViT fine tuned model parameters

Parameter	Dimension	Activation functions
Embedding Layer	xx,1024	N/A
Dense Neural Network	1024,64	ReLu
Drop Out Layer	N/A	Factor = 0.5
Dense Neural network	64, NumberofClasses	softmax

Algorithm Run Setup: Few Shot Prompting Experiment SetUp

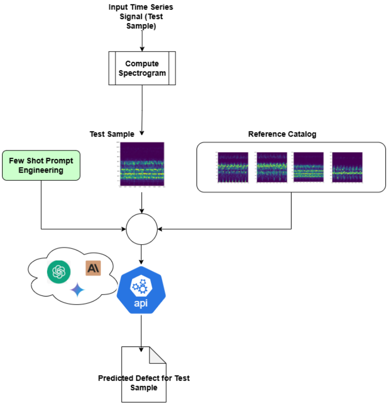


Figure: Few Shot Prompting Experiment SetUp

Algorithm Run Setup: API-Few Shot Prompting

```
{"role": "user",
"content": [
  {"type": "text",
  "text": "You are an expert scientific spectogram analyzer. Provided below are examples
    with tag ##EXAMPLE## are spectrogram corresponding to different defect classes with a
    tag ##DEFECT CLASS ##. Given a plot of spectrogram images with x axis in time and y
    axis in frequeny. Amplitude of the signals are color coded in standard Viridis scale.
    Extract key spectrogram information such as peak frequencies, peak amplitudes,
    frequency band characteristics, any harmonic patterns or any other relevant time
    spectrogram features.Dont hallucinate. Extract information strictly from the plot and
    associate it with the defect class"},
    {"type": "text",
    "text": "##EXAMPLE##  ##DEFECT CLASS: ##ACCEPT"},
    {"type": "image", "source": "#base64RefImage"},
    {"type": "text", "text": "##EXAMPLE##  ##DEFECT CLASS: ##OUTER DENT"},
    {"type": "image","source": "#base64RefImage"},
    {"type": "text", "text": "##EXAMPLE##  ##DEFECT CLASS: ##NEEDLE REJECT"},
    {"type": "image","source": "#base64RefImage"},
    {"type": "text", "text": "##EXAMPLE##  ##DEFECT CLASS: ##RACEWAY DENT"},
    {"type": "image","source": "#base64RefImage"},

    {"type": "text",
    "text": "Now, given a new spectrogram , analyze it considering factors such as
      frequency patterns,intensity and time variations. Focus solely on the patterns
      presented in the spectrogram. Do not let any assumptions or environmental settings
      influence your decision. Your task is to determine which of  the ##EXAMPLE##
      classes the new spectrogram most closely resembles . Your response must contain only
      the exact name of the class. Predicted Class:"},
    {"type": "image","source": "base64ImageTest"}
]}
```

Figure: API Few Shot Prompting

Study Results: Ablation Study Spectrogram Plots

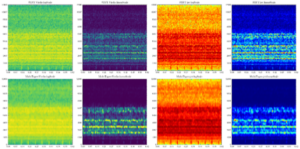


Figure: Spectrogram Ablation Study options

Table: Spectrogram Ablation study Results

Param	Few Shot F1
STFT Virdis Log Ampl	0.56
STFT Virdis Linear Ampl	0.74
STFT Jet Log Ampl	0.58
STFT Jet Linear Ampl	0.82
MTFT Virdis Log Ampl	0.54
MTFT Virdis Linear Ampl	0.88
MTFT Jet Log Ampl	0.52
MTFT Jet Linear Ampl	0.86

Spectrogram Type Chosen

Multitaper VIRDIS colormap Linear Amptd scale

Study Results: Experiment Run Results - Baseline Performances

Table: Few Shot Human performance study

Candidate	Few Shot F1 macro
Human expert 1	0.78
Human expert 2	0.86
Human expert 3	0.82
Average Score	0.82

Table: ViT model performance

Precision macro	Recall macro	F1 score macro
0.95	0.97	0.98

Baseline Scores

F1 score - Human Expert score is 0.82 and GPT4o is 0.90

Study Results: Experiment Run Results - One shot Prompting Results

Table: One Shot Performance

Model	Precision macro	Recall macro	F1 macro
GPT-4o	0.90	0.90	0.90
GPT-4o-mini	0.76	0.53	0.45
Claude-3-5-Sonnet	0.86	0.86	0.86
Claude-3-Opus	0.24	0.24	0.24
Gemini-1.5-Flash	0.54	0.34	0.25
Gemini-1.5-Pro	0.39	0.45	0.40

One Shot Scores

GPT-40, Claude-3-5-Sonnet and Gemini 1.5 Pro only are chosen

Study Results: Experiment Run Results - Few shot Prompting Results

Table: Few shot (K=2) Shot Performance

Model	Precision macro	Recall macro	F1 macro
GPT-4o	0.95	0.95	0.95
Claude-3-5-Sonnet	0.94	0.94	0.94
Gemini-1.5-Pro	0.42	0.48	0.45

Few Shot Scores

GPT-4o Top Ranked leader board with F1 score of 0.95

Study Results: Experiment Run Results - Few shot Prompting Confusion Matrix

Table: GPT-4o-Confusion Matrix

		Predicted			
		ACPT	DENT	NR	RD
Actual	ACPT	90	10	0	1
	DENT	2	97	1	0
	NR	5	0	94	1
	RD	0	0	0	100

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Result Summary

The key results are :

1. Data centric study of vibration data for needle bearings
2. Proven Vision Language Models are Few Shot Learners of Spectrograms
3. Achieving Performance better than Human Experts and Very close to SOTA model
4. Tremendous Potential for several industrial applications for analyzing or decision making with spectrograms

Result Summary

Limitations of the Study and Future Research :

1. Tested only selected closed source VLM models (Study done before DeepSeek Release)
2. Open Source VLM models are not studied
3. Token size limitations reduces spectrogram resolution and also few shot options
4. Assesment done only for limited 4 class study. More faulty and higher number of classses study needed
5. Fine Tuning is out of scope of the study. Lot of potential for improving performance with fine tuning.

References

1. Dixit, S., Heller, L. M., and Donahue, C., Vision Language Models Are Few-Shot Audio Spectrogram Classifiers arXiv, 2024. [Online].
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3. Neupane and Seok, J. Bearing Fault Detection and Diagnosis Using Case Western Reserve University Dataset with Deep Learning Approaches A Review." *IEEE Access*, 8, 93155-93178 (2020).
4. Zim, Abid Hasan, Ashraf, Aeyan, Iqbal, Aquib, Malik, Asad, and Kuribayashi, Minoru. A Vision Transformer-Based Approach to Bearing Fault Classification via Vibration Signals. *arXiv* (2022).

Thank You!
Questions are welcome!!!